

Ch 2. Rigid Body Motion.

Def 2.1 A mapping $g: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a rigid body transformation if

① $\|g(p) - g(q)\| = \|p - q\|, \forall p, q \in \mathbb{R}^3$. ② $g_*(v \times w) = g_*(v) \times g_*(w) \quad \forall v, w \in \mathbb{R}^3$.

Diagram: A point p is mapped to $g(p)$. A vector $v = g - p$ is mapped to $g_*(v) = g(g) - g(p)$.
 $g_*(v) = g(g) - g(p)$
 $g_*(0) = 0$
 $g_*(-v) = -g_*(v)$
 $g_*(v) = g(v) - g(0)$
 $\|g_*(p) - g_*(q)\| = \|p - q\|$

$$\forall v_1, v_2 \in \mathbb{R}^3, \quad v_1^T v_2 = \frac{1}{4} (\|v_1 + v_2\|^2 - \|v_1 - v_2\|^2)$$

$$= \frac{1}{4} (\|g_*(v_1) + g_*(v_2)\|^2 - \|g_*(v_1) - g_*(v_2)\|^2) = g_*(v_1)^T g_*(v_2)$$

* Rotation

Space of rotation matrices $SO(n)$.

~~Def~~ $SO(n) := \{R \in \mathbb{R}^{n \times n} : R^T R = I, \det R = 1\} \rightarrow \text{group}$

$SO(3)$: rotation group of $\mathbb{R}^3$.

For $a \in \mathbb{R}^3$, define $(a)^\wedge = \hat{a} := \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix} \rightarrow \hat{a}^T = -\hat{a} : \text{skew-symmetric}$. Then $a \times b = (a)^\wedge b$.

Lemma 2.1 $R \in SO(3)$, $v, w \in \mathbb{R}^3$, then

$$R(v \times w) = Rv \times Rw, \quad R(w)^\wedge R^T = (Rw)^\wedge$$

$$R = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}, \quad (Rv \times Rw)_i = \begin{vmatrix} r_2 \cdot v & r_3 \cdot v \\ r_2 \cdot w & r_3 \cdot w \end{vmatrix} = (r_2 \times r_3) \cdot (v \times w) = r_1 \cdot (v \times w) = (R(v \times w))_i$$

Prop 2.2 $R \in SO(3)$ is a rigid body transformation.

$\therefore \|Rp - Rq\| = \|p - q\|$

$$\|Rp - Rq\|^2 = (p - q)^T R^T R (p - q) = (p - q)^T (p - q) = \|p - q\|^2$$

* Exponential coordinates

Diagram: A cylinder with a vector $g(0)$ and a rotated vector $g(t)$. A rotation arrow is labeled ω .
 $\dot{g}(t) = \omega \times g(t) = \hat{\omega} g(t)$

$$\Rightarrow g(t) = e^{\hat{\omega} t} g(0), \quad e^{\hat{\omega} t} = I + \hat{\omega} t + \frac{(\hat{\omega} t)^2}{2!} + \dots$$

$\hat{\omega}^T = -\hat{\omega}$, $so(n) := \{S \in \mathbb{R}^{n \times n} : S^T = -S\}$. Rotation $R(\omega, \theta) = e^{\hat{\omega} \theta}$

Lemma 2.3 $\hat{a} \in so(3)$, then $\hat{a}^2 = aa^T - \|a\|^2 I$, $\hat{a}^3 = -\|a\|^2 \hat{a}$ (direct calculation)

If $\|\omega\| = 1$, then $e^{\hat{\omega} \theta} = I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) \hat{\omega} + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots\right) \hat{\omega}^2$
 $= I + \hat{\omega} \sin \theta + \hat{\omega}^2 (1 - \cos \theta) : \text{Rodrigues' formula}$

~~$\hat{\omega} = \omega^\wedge$~~

$\hat{\omega}^2 = -\hat{\omega}$

Prop 2.4 $\hat{\omega} \in \mathfrak{so}(3)$, $\theta \in \mathbb{R} \Rightarrow e^{\hat{\omega}\theta} \in SO(3)$ WLOG, $\|\omega\|=1$.

$$(e^{\hat{\omega}\theta})^{-1} = e^{-\hat{\omega}\theta} = e^{\hat{\omega}^T\theta} = (e^{\hat{\omega}\theta})^T$$

$$\therefore (I + \hat{\omega} \sin \theta + \hat{\omega}^2 (1 - \cos \theta)) (I - \hat{\omega} \sin \theta + \hat{\omega}^2 (1 - \cos \theta))$$

$$= (I + \hat{\omega}^2 (1 - \cos \theta))^2 - \hat{\omega}^2 \sin^2 \theta = I$$

$$I + 2\hat{\omega}^2(1 - \cos \theta) + \hat{\omega}^4(1 - 2\cos \theta + \cos^2 \theta) = I + 2\hat{\omega}^2(1 - \cos \theta) - \hat{\omega}^2(1 - 2\cos \theta + \cos^2 \theta)$$

$$= I + \hat{\omega}^2 - \hat{\omega}^2 \cos^2 \theta$$

$$\det e^{\hat{\omega}\theta} = 1?$$

$$\theta=0 \Rightarrow \det e^{\hat{\omega}\theta} = 1, \quad \det \circ \exp: \text{continuous}$$

□

Prop 2.5 $\forall R \in SO(3)$, $\exists \omega \in \mathbb{R}^3$, $\|\omega\|=1$ & $\theta \in \mathbb{R}$ s.t. $R = \exp(\hat{\omega}\theta)$.

pf) $\det R = 1$, $\|Rv\| = \|v\| \Rightarrow |\lambda| = 1$ $\{1, \cancel{e^{i\theta}}, e^{-i\theta}\}$ or $\{-1, e^{i\theta}, e^{-i\theta}\}$

$$\therefore \{1, e^{i\theta}, e^{-i\theta}\}$$

$$\cancel{1 \cdot e^{i\theta} \cdot e^{-i\theta} = 1}$$

$$1 \cdot e^{i\theta} \cdot e^{-i\theta} = 1$$

$$\Rightarrow \text{tr } R = \lambda_1 + \lambda_2 + \lambda_3, \quad -1 \leq \text{tr } R \leq 3$$

$$\theta := \cos^{-1}\left(\frac{\text{tr } R - 1}{2}\right), \quad \omega = \frac{1}{2\sin \theta} \begin{pmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{pmatrix} \Rightarrow R = \exp(\hat{\omega}\theta) \quad \square$$

$\omega\theta \in \mathbb{R}^3$: exponential coordinates of R .

Thm 2.6 Any orientation $R \in SO(3)$ is equiv. to a rotation about a fixed axis $\omega \in \mathbb{R}^3$ through an angle $\theta \in [0, 2\pi)$.

Other representations?

1. Euler angles.

$$(d, p, r) \rightarrow SO(3)$$

rotate about z-axis by angle d ,

" y-axis " p .

" z-axis " r .

$$R = R_z(d) R_y(p) R_z(r)$$

then $(d, p, r) \rightarrow SO(3)$ is surjective.

ZYZ, ZYX, YZX...

2. Quaternions.

$$(q_0, \vec{q}), \quad \vec{q} = (q_1, q_2, q_3)$$

$$Q = q_0 + q_1 i + q_2 j + q_3 k, \quad \|Q\|^2 = q_0^2 + q_1^2 + q_2^2 + q_3^2$$

$$R = \exp(\hat{\omega}\theta) \Rightarrow Q = \left(\cos\left(\frac{\theta}{2}\right), \omega \sin\left(\frac{\theta}{2}\right)\right) \in \{Q : \|Q\|=1\}$$

* Rigid Motion in $\mathbb{R}^3$.

Given $p \in \mathbb{R}^3$, $R \in SO(3)$, we can define $g(g) = p + Rg = \text{rigid motion}$.

$$SE(3) := \{(p, R) : p \in \mathbb{R}^3, R \in SO(3)\} = \mathbb{R}^3 \times SO(3).$$

$$SE(n) := \mathbb{R}^n \times SO(n).$$

$$g_*(v) = Rv.$$

$\bar{g}$: ~~homogeneous~~ homogeneous representation of g .

For $g \in (p, R) \in SE(3)$, ~~define~~ $\bar{g} := \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 4}$. $\begin{pmatrix} R_1 & p_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_2 & p_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} R_1 R_2 & R_1 p_2 + p_1 \\ 0 & 1 \end{pmatrix}$

Then $SE(3)$ forms a group: ① $g_1, g_2 \in SE(3) \Rightarrow g_1 g_2 \in SE(3)$.

$$\textcircled{2} I \in SE(3)$$

$$\textcircled{3} \bar{g}^{-1} = \begin{pmatrix} R^T & -R^T p \\ 0 & 1 \end{pmatrix} \in SE(3), g^{-1} = (-R^T p, R^T)$$

Prop 2.7 $\forall g \in SE(3)$ is a rigid body transformation

$$\therefore \|g g_1 - g g_2\| = \|R g_1 - R g_2\| = \|g_1 - g_2\|, g_* v \times g_* w = Rv \times Rw = R(v \times w)$$

* exponential coordinates.

$$se(3) := \{(v, \hat{\omega}) : v \in \mathbb{R}^3, \hat{\omega} \in so(3)\}, \hat{\xi} \in se(3), \hat{\xi} = \begin{pmatrix} \hat{\omega} & v \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{4 \times 4}$$

$$v, \wedge : \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} v \\ w \end{bmatrix}, \begin{bmatrix} v \\ w \end{bmatrix}^\wedge = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix}$$

Prop 2.8 $\forall \hat{\xi} \in se(3), \theta \in \mathbb{R}, e^{\hat{\xi}\theta} \in SE(3)$.

pf) $\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix}$. If $\omega = 0$, then $\hat{\xi}^\wedge = 0 \Rightarrow \exp(\hat{\xi}\theta) = I + \hat{\xi}\theta = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix} \in SE(3)$.

($\omega \neq 0$). Assume $\|\omega\| = 1$.

$$\hat{\xi}^\wedge \hat{\xi} := g^{-1} \hat{\xi} g = \begin{bmatrix} I & -\omega \times v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I & \omega \times v \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \hat{\omega} & \omega \omega^T v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \hat{\omega} & (\omega^T v) \omega \\ 0 & 0 \end{bmatrix}$$

$$e^{\hat{\xi}\theta} = \begin{pmatrix} e^{\hat{\omega}\theta} & (\omega^T v) \omega \theta \\ 0 & 1 \end{pmatrix} \because (\hat{\xi}^\wedge)^2 = \begin{pmatrix} \hat{\omega}^2 & 0 \\ 0 & 0 \end{pmatrix}, (\hat{\xi}^\wedge)^3 = \begin{pmatrix} \hat{\omega}^3 & 0 \\ 0 & 0 \end{pmatrix} (\because \hat{\omega}\omega = \omega \times \omega = 0)$$

$$= I + \hat{\xi}\theta + \frac{1}{2!} \hat{\xi}^2 \theta^2 + \dots$$

$$\Rightarrow e^{\hat{\xi}\theta} = e^{g(\hat{\xi}\theta)g^{-1}} = g e^{\hat{\xi}\theta} g^{-1} = \begin{pmatrix} I & \omega \times v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e^{\hat{\omega}\theta} & (\omega^T v) \omega \theta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} I & -\omega \times v \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} e^{\hat{\omega}\theta} & (\omega^T v) \omega \theta + \omega \times v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} I & -\omega \times v \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})(\omega \times v) + \omega \omega^T v \theta \\ 0 & 1 \end{pmatrix} \in SE(3)$$

Prop 2.7 $\forall g \in SE(3), \exists \hat{z} \in se(3), \theta \in \mathbb{R} \text{ s.t. } g = e^{\hat{z}\theta}. \hat{z} = \begin{pmatrix} \hat{\omega} & v \\ 0 & 0 \end{pmatrix}$

f) i) $(R, p) = (I, 0) : \theta = 0$

ii) $R = I, p \neq 0. \hat{z} = \begin{pmatrix} 0 & \frac{p}{\|p\|} \\ 0 & 0 \end{pmatrix}, \theta = \|p\| \Rightarrow e^{\hat{z}\theta} = I + \hat{z}\theta = (I, p) = g$

iii) $R \neq I$

$$e^{\hat{z}\theta} = \begin{pmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})(\omega \times v) + \omega \omega^T v \theta \\ 0 & 1 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix}$$

Use Prop 2.5 to find $\hat{\omega}, \theta$ s.t. $e^{\hat{\omega}\theta} = R$

$A = (I - e^{\hat{\omega}\theta})\hat{\omega} + \omega \omega^T \theta$: nonsingular? $\langle \omega \rangle = \text{span}\{\omega\}$

$\text{Im } \hat{\omega} = \langle \omega \rangle^\perp, \ker \hat{\omega} = \langle \omega \rangle. P_{\langle \omega \rangle} = \frac{\omega \omega^T}{\omega^T \omega}, P_{\langle \omega \rangle^\perp} = I - \frac{\omega \omega^T}{\omega^T \omega}$

$\ker(I - e^{\hat{\omega}\theta}) = \langle \omega \rangle \rightarrow I - e^{\hat{\omega}\theta} : \langle \omega \rangle^\perp \rightarrow \langle \omega \rangle^\perp$ is bijective.

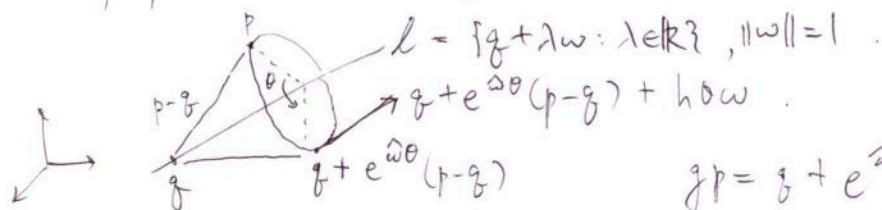
$\therefore e^{\hat{\omega}\theta} = \cos \theta I + (1 - \cos \theta) \frac{\omega \omega^T}{\omega^T \omega} + \sin \theta \hat{\omega} : \langle \omega \rangle^\perp \rightarrow \langle \omega \rangle^\perp$

$A|_{\langle \omega \rangle} = \theta I$: invertible. $A|_{\langle \omega \rangle^\perp} = (I - e^{\hat{\omega}\theta})\hat{\omega} : \text{bijective on } \langle \omega \rangle^\perp$

$A(\langle \omega \rangle) \subset \langle \omega \rangle, A(\langle \omega \rangle^\perp) \subset \langle \omega \rangle^\perp, A|_{\langle \omega \rangle}, A|_{\langle \omega \rangle^\perp} : \text{invertible. } \mathbb{R}^3 = \langle \omega \rangle \oplus \langle \omega \rangle^\perp$

$\therefore A$ is invertible. $\square$

Def A screw (l, h, M) represents rotation by an amount $\theta = M$ about the axis l followed by translation by an amount $h\theta$ parallel to axis l . If $h = \infty$, then only pure translation by distance M .



$$gp = g + e^{\hat{\omega}\theta}(p - g) + h\theta\omega$$

$$g[p] = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})g + h\theta\omega \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix}$$

$$e^{\hat{z}\theta} = \begin{pmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})(\omega \times v) + \omega \omega^T v \theta \\ 0 & 1 \end{pmatrix}$$

screw $\longleftrightarrow$ twist $(\xi = (\omega, v), \theta)$

$$(l, h, M) \Rightarrow \omega \Leftrightarrow v = -\omega \times g + h\omega, \hat{z} = \begin{pmatrix} \hat{\omega} & v \\ 0 & 0 \end{pmatrix} \theta = M$$

$l = \{g + \lambda\omega : \lambda \in \mathbb{R}\}, \|\omega\| = 1$. If $h = \infty, \hat{z} = \begin{pmatrix} \hat{\omega} & \omega \\ 0 & 0 \end{pmatrix}$

$$(\xi = (\omega, v), \theta) \rightarrow \begin{cases} h = \frac{\omega^T v}{\|\omega\|^2} \\ l = \begin{cases} \left\{ \frac{\omega \times v}{\|\omega\|^2} + \lambda\omega : \lambda \in \mathbb{R} \right\} & \omega \neq 0 \\ \{0 + \lambda v : \lambda \in \mathbb{R}\} & \omega = 0 \end{cases} \\ M = \begin{cases} \|\omega\| & \omega \neq 0 \\ \|v\| & \omega = 0 \end{cases} \end{cases}$$

Thm 2.11 Every rigid body motion can be realized by a rotation about an axis combined with a translation parallel to that axis.